Lab 1: Requirements Engineering & UML Use-Case Modelling

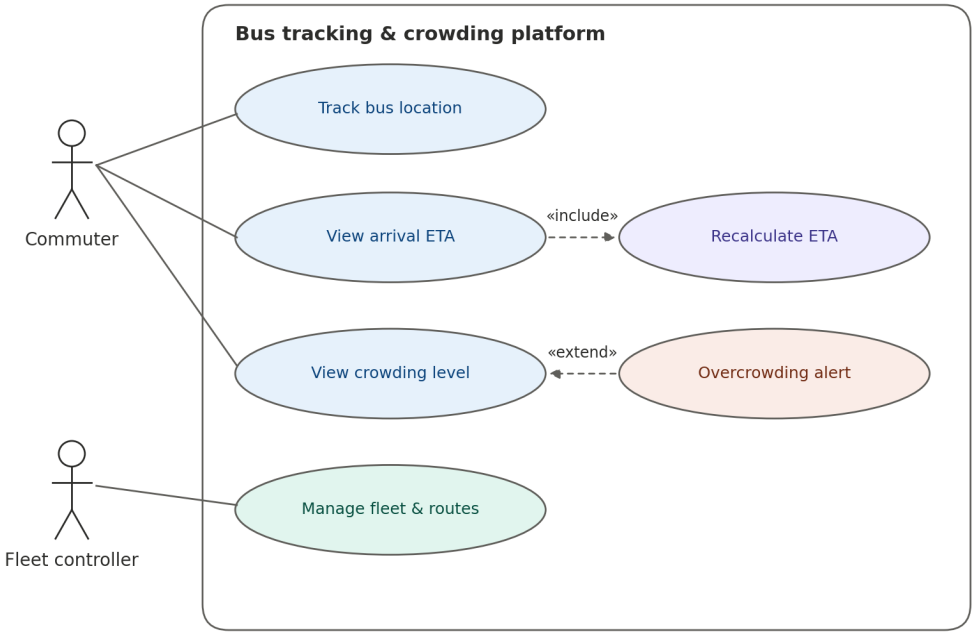
Problem Statement #24 — Public Bus Live Tracking & Crowding Estimator

Domain: Smart Cities, Transport & Logistics | Actors: Commuter, Fleet Controller

1. Requirements Table

ID	Type	Priority	Description	Acceptance criteria	Rationale
FR-001	Functional	High	System shall ingest GPS telemetry every 5 seconds per vehicle and recompute ETA for all downstream stops on the route.	Pass: ETA decreases monotonically as the bus approaches. Fail: ETA increases while approaching.	Commuters need continuously updating, accurate arrival predictions.
FR-002	Functional	High	System shall compute a crowding level (Low/Medium/High) per active bus from ticketing/boarding sensor counts, refreshed at least once per minute.	Pass: crowding level matches sensor counts within defined thresholds. Fail: stale level shown after a material passenger-count change.	Lets commuters make informed boarding decisions and avoid overcrowded buses.
FR-003	Functional	Medium	Commuter app shall show real-time bus positions on a map, updated at the same 5-second cadence as GPS ingestion.	Pass: marker is within 50m of actual GPS reading. Fail: marker lags by more than one ingestion cycle.	Visual tracking builds commuter trust and reduces wait-time uncertainty.
FR-004	Functional	High	System shall provide fleet controllers a dashboard of all active vehicles (position, ETA adherence, crowding), with automatic alerts when a vehicle is delayed past a configurable threshold.	Pass: a vehicle delayed >5 min past scheduled ETA is flagged within 30s of breach. Fail: delayed vehicle not flagged.	Enables proactive fleet management and service recovery.
FR-005	Functional	Medium	Fleet controllers shall be able to add, edit, or deactivate routes and stops, with changes reflected in ETA/crowding calculations within 2 minutes.	Pass: a newly added stop appears in ETA calculations within 2 minutes. Fail: stale route data used after an edit.	Keeps the system aligned with real-world diversions and new stops.
NFR-001	Performance	High	Tracking engine shall handle telemetry from up to 1,000 concurrently active vehicles without degradation.	Pass: benchmark confirms sub-2-second ETA recompute latency at 1,000 simulated vehicles. Fail: latency exceeds 2s or data loss occurs.	Platform must support city-scale fleets during peak hours.
NFR-002	Security	High	System shall encrypt GPS telemetry and crowding data in transit and at rest, and restrict the fleet-controller dashboard to authenticated, authorized users.	Pass: penetration test confirms TLS in transit, AES-256 at rest, and rejected unauthorized access. Fail: any unencrypted channel or successful unauthorized access.	Protects sensitive location and ridership data from interception or misuse.

2. UML Use-Case Diagram



Actors: Commuter, Fleet controller. «include»: View arrival ETA includes Recalculate ETA. «extend»: Overcrowding alert extends View crowding level.

3. Use-Case Flow Specification — View Arrival ETA

Actor

Commuter

Preconditions

Commuter has the app open with location access granted; at least one active bus is running on a route serving the selected stop.

Postconditions

Commuter sees a current ETA for the selected stop; the query is logged for analytics.

Main Success Scenario

1 Commuter selects a stop in the app.
2 System retrieves the buses en route to that stop.
3 System invokes «include» Recalculate ETA to compute an up-to-date estimate from the latest GPS telemetry.
4 System displays ETA(s) for each approaching bus at the stop.
5 Commuter views the ETA and decides whether to head to the stop.

Alternate Flow A1 — No GPS Signal from Bus

1a At step 3, telemetry for a bus hasn't been received in the last 30 seconds.
2a System marks that bus's ETA as “unavailable” rather than computing a value.
3a System shows the last-known position with a “signal lost” indicator.
4a Flow resumes at step 4 for the remaining buses with current telemetry.